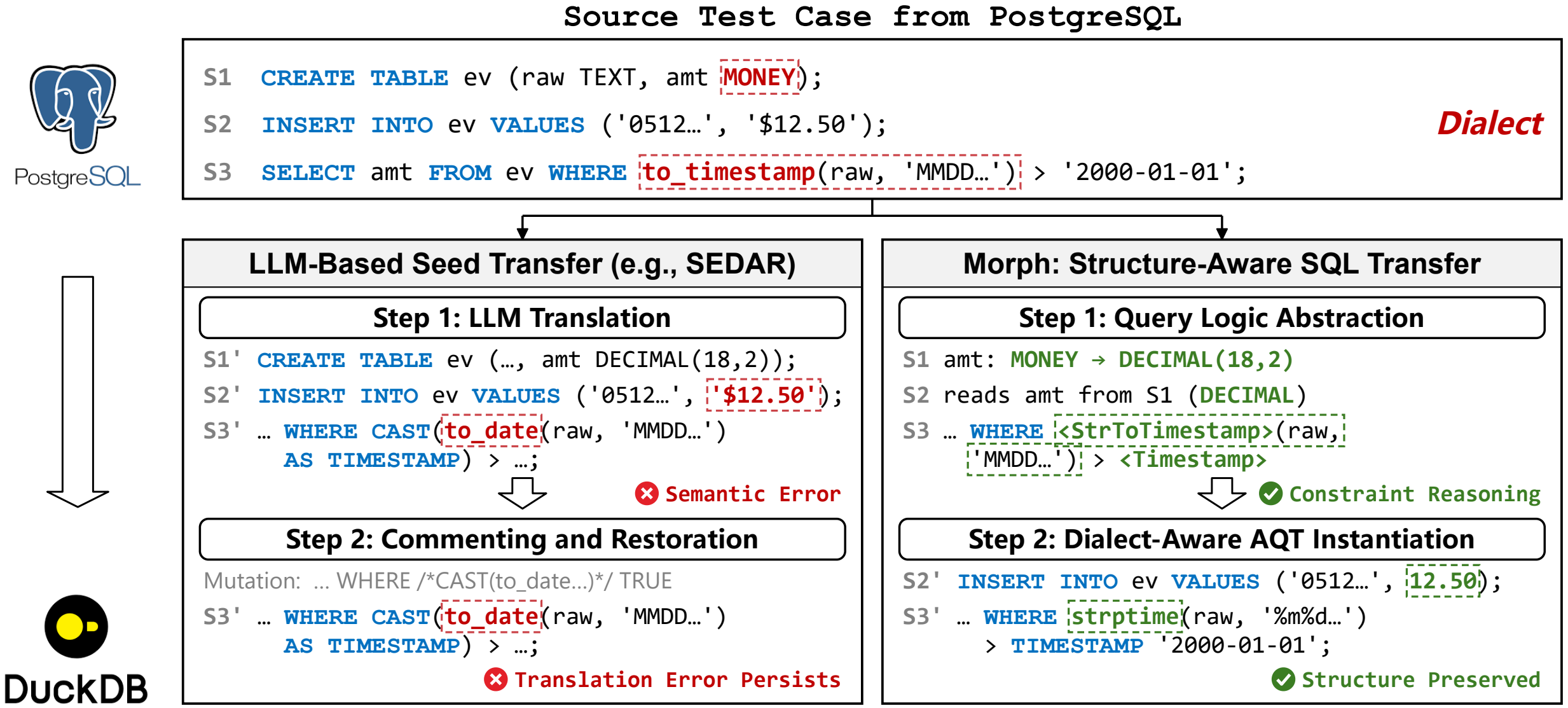

Source Test Case from PostgreSQL
PostgreSQL
S1 CREATE TABLE ev (raw TEXT, amt MONEY);
S2 INSERT INTO ev VALUES ('0512…', '$12.50');
S3 SELECT amt FROM ev WHERE to_timestamp(raw, 'MMDD…') > '2000-01-01';
Dialect
LLM-Based Seed Transfer (e.g., SEDAR)
Step 1: LLM Translation
S1' CREATE TABLE ev (…, amt DECIMAL(18,2));
S2' INSERT INTO ev VALUES ('0512…', '$12.50');
S3' … WHERE CAST(to_date(raw, 'MMDD…') AS TIMESTAMP) > …;
Semantic Error
Step 2: Commenting and Restoration
Mutation: … WHERE /*CAST(to_date…)*/ TRUE
S3' … WHERE CAST(to_date(raw, 'MMDD…') AS TIMESTAMP) > …;
Translation Error Persists
Morph: Structure-Aware SQL Transfer
Step 1: Query Logic Abstraction
S1 amt: MONEY → DECIMAL(18,2)
S2 reads amt from S1 (DECIMAL)
S3 … WHERE <StrToTimestamp>(raw, 'MMDD…') > <Timestamp>
Constraint Reasoning
Step 2: Dialect-Aware AQT Instantiation
S2' INSERT INTO ev VALUES ('0512…', 12.50);
S3' … WHERE strptime(raw, '%m%d…') > TIMESTAMP '2000-01-01';
Structure Preserved
DuckDB